Leveraging Cooperative Artificial Intelligence (AI) Systems for Analysis of Qualitative Assessment Data: A Case Study

-- OR --

When AI Creates More Work: Managing “Code Explosion” in Large Qualitative Datasets

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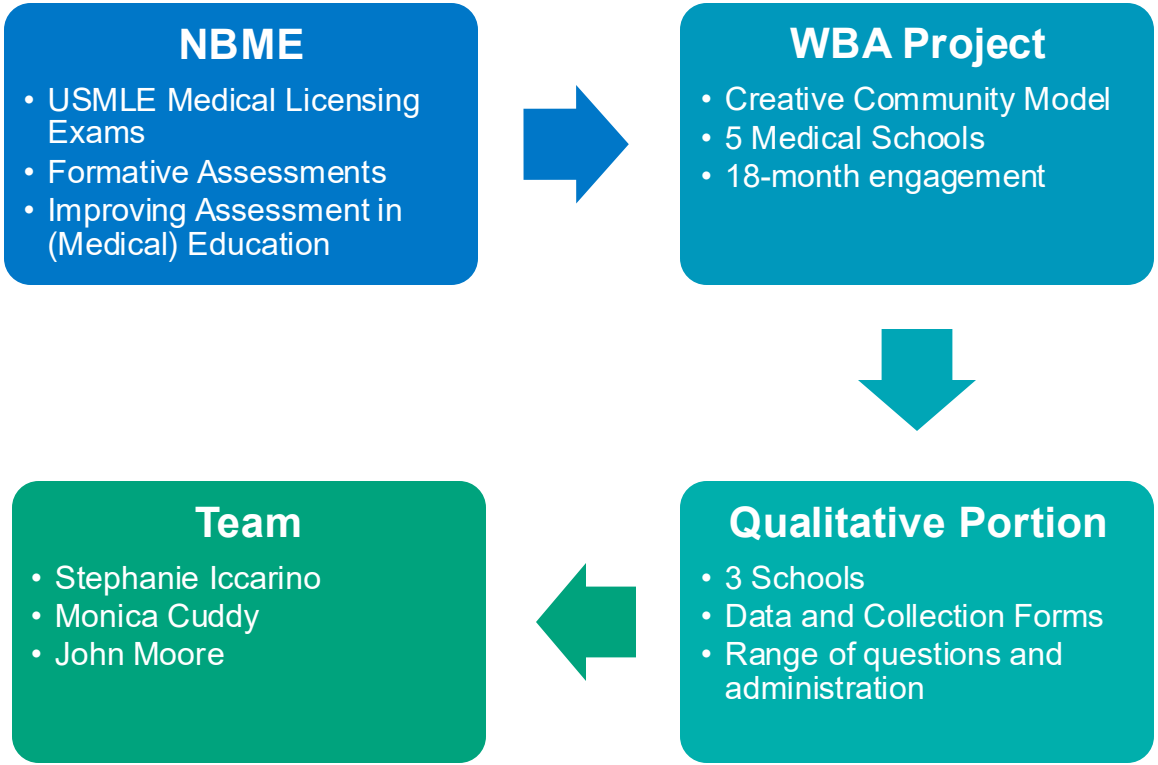
Agenda

- Introductions / Presentation Goals
- Background (NBME, WBA Project)
- Discussion I
- Qualitative Component: AI, Process, and Challenges
- Discussion II
- Findings (Project, Process)
- Lessons Learned
- Discussion III / Questions

Session Goals

- Understand the benefits and risks of using AI in analyzing qualitative data.
- Appraise the effectiveness for Human in the Loop and Multiple AI Systems to avoid “Code Explosion.”
- Evaluate Code Output and Edit Prompts for better results.
- Formulate a plan for investing the use of AI for qualitative analysis at your institution.

Introductions and Background



Why WBA?

Why Workplace-Based Assessment Data?

- Challenging Area for Schools
- NBME Prioritized Areas: Clinical Reasoning, Communications, Teamwork (HSS)
- NBME’s potential for contribution

What we received

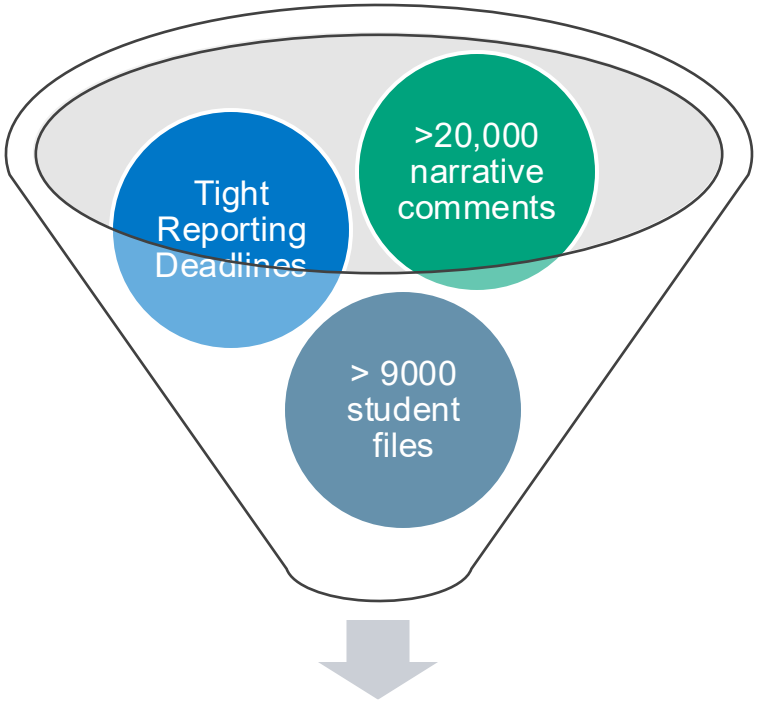
- Four years of data from WBAs
- Approximately 2.5 million lines of data (Qual and Quant)
- 3 Schools contributed Qualitative Data

	School 1	School 2	School 3
Average Class Size	81	625	100
Students w/ Comment	306	1398	471
Comments	2240	23,182	15,202
Mean (SD) Comments/Student	3 (.7)	17 (12)	32 (12)

- This presentation is a case study of the work we did on school 2.

Have you ever experienced a time when you had too much qualitative data to reasonably analyze?

How can large volumes of narrative data be reduced to manageable levels?



Manual Coding?

Sampling?

Automation?

Decision #1: Use AI-Assisted Coding to Reduce Workload and Increase Coverage

- ☑ Met organizational and policy approvals for sharing data with AI feature
- ☑ Explore Computer-Assisted Qualitative Data Analysis Software (CAQDAS) AI Coding Functionality and Requirements



Problem Solved!

Result: “Code Explosion” Problem

“Feedback Application”

“Feedback”

“Integration of Feedback”

“Feedback Integration”

“Feedback Incorporation”

“Feedback Utilization”

“Feedback Adaptation”

“Feedback Adaptability”

“Feedback Appreciation”

“Feedback Openness”

“Receptive to Feedback”

“Openness to Feedback”

“Feedback Implementation”

“Feedback Receptiveness”

“Feedback Responsiveness”

“Feedback Reception”

Result: “Code Explosion” Problem

“Feedback Application”

“Feedback”

“**Integration** of Feedback”

“Feedback **Integration**”

“Feedback Incorporation”

“Feedback Utilization”

“Feedback **Adaptation**”

“Feedback Appreciation”

“Feedback **Adaptability**”

“Feedback **Openness**”

“**Receptive** to Feedback”

“**Openness** to Feedback”

“Feedback Implementation”

“Feedback **Receptiveness**”

“Feedback Responsiveness”

At this point, we had simply traded one human-intensive task for another.

“Feedback **Reception**”

Decision #2: Pivot to Generative AI Platform for Code Reduction

Qualitative Software

- ✓ User-friendly interface for coding/memorandums
- ✗ Produces too many codes
- ✗ Minimal control over prompting

Generative AI Platform

- ✗ Lacks user interface for qualitative coding
- ✓ Supports prompting control

Goal: Accurately Represent Data while Minimizing Code List for Human Review

Decision #3 – Strategically Compose Prompts to Increase Likelihood of Expected Output

Initial Prompt Feature	Example from Initial Prompt
Task Description	"Re-categorize each entry (cell) into a newly-created list containing least amount of sensible categories."
Formatting Guidelines	"Categorize cells that are similar in name regardless of plurality/spelling (e.g., feedback and fedback -> category name = Feedback)"
Initial Codebook	**Category**: Clinical Knowledge **Description**: Identified clinical and/or relevant knowledge/lack thereof

Initial Prompt

Your task:
Re-categorize each entry (cell) into a newly-created list containing least amount of sensible categories

Guidelines:

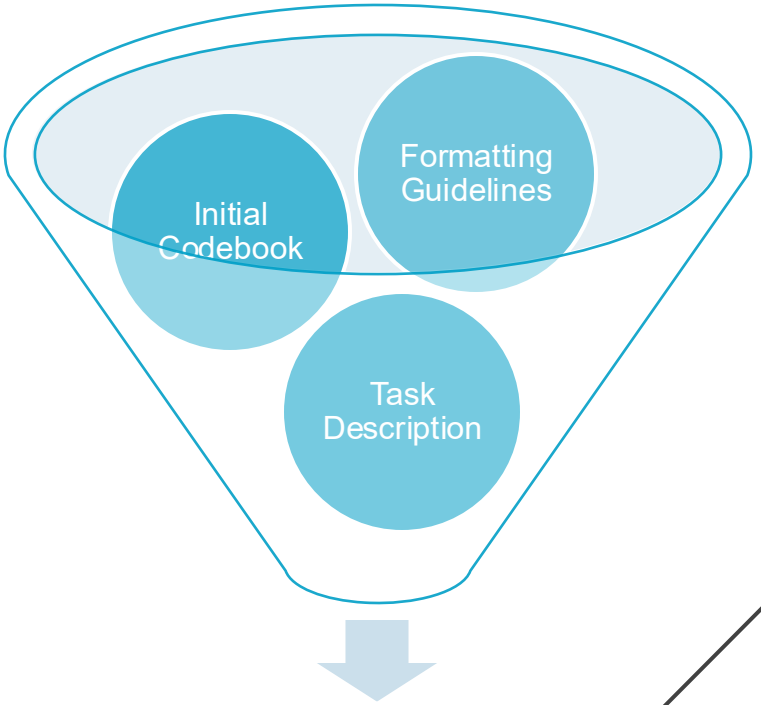
- Base categories on the exact similarity current text in each cell: (e.g., team reception -> category name = Teamwork)
- Categorize cells that are similar in name regardless of plurality/spelling (e.g., feedback and fedback -> category name = Feedback)
- Categorize cells that contain similar thematic meaning (e.g., “loves to learn,” “willing to try new things” -> category name = Learning Approaches
- Use the table below to inform categorizations, but also you may create new categories, as well

Category	Description
Clinical Knowledge	Identified Clinical and/or relevant knowledge/lack thereof

- CRITICAL – every cell should be categorized
- Return output as an Excel spreadsheet

Generate the response now

Decision #3 – Strategically Compose Prompts to Increase Likelihood of Expected Output



Original CAQDAS-generated code	Platform-generated code category
Original $Code_1$	Generated Code $Category_1$
Original $Code_2$	Generated Code $Category_2$

How Did We Evaluate Prompt Output?

Accuracy

Parsimony

Representativeness

How Did We Evaluate Prompt Output?

Accuracy: Thematic similarity to codebook



Codebook:
"Collaborate"
→ Teamwork

Output:
"Collaboration" → Teamwork



Codebook:
"Improve"
→ Feedback and Improvement

Output:
"Clinical Improvement"
→ Clinical Skills and Behaviors



Prompt Revision(s):

- Include more specific examples for each code

Example:

Category	Description	Example Words
Clinical Knowledge	Identified **and/or applied** clinical, **medical** and/or relevant knowledge/lack thereof **(about patient, preexisting knowledge)**	Knowledge, principles

Inconsistencies required additional specificity either to the prompt instructions or to the codebook itself

How Did We Evaluate Prompt Output?

Parsimony: Reduce number of codes for feasible human review



N = 20 code categories



N = 100 code categories



Add an explicit "category cap" to the prompt

Example:

Guidelines:
Cap on total categories:
Maximum of 20 categories (including base categories)

Too many categories made formal review infeasible, even when it created meaningful groupings at face value

How Did We Evaluate Prompt Output?

Representativeness: Code categories are internally meaningful and consistent



Other category = 0% of codes

Non- “Other” category = 8% of codes



“Other” category = 91% of codes



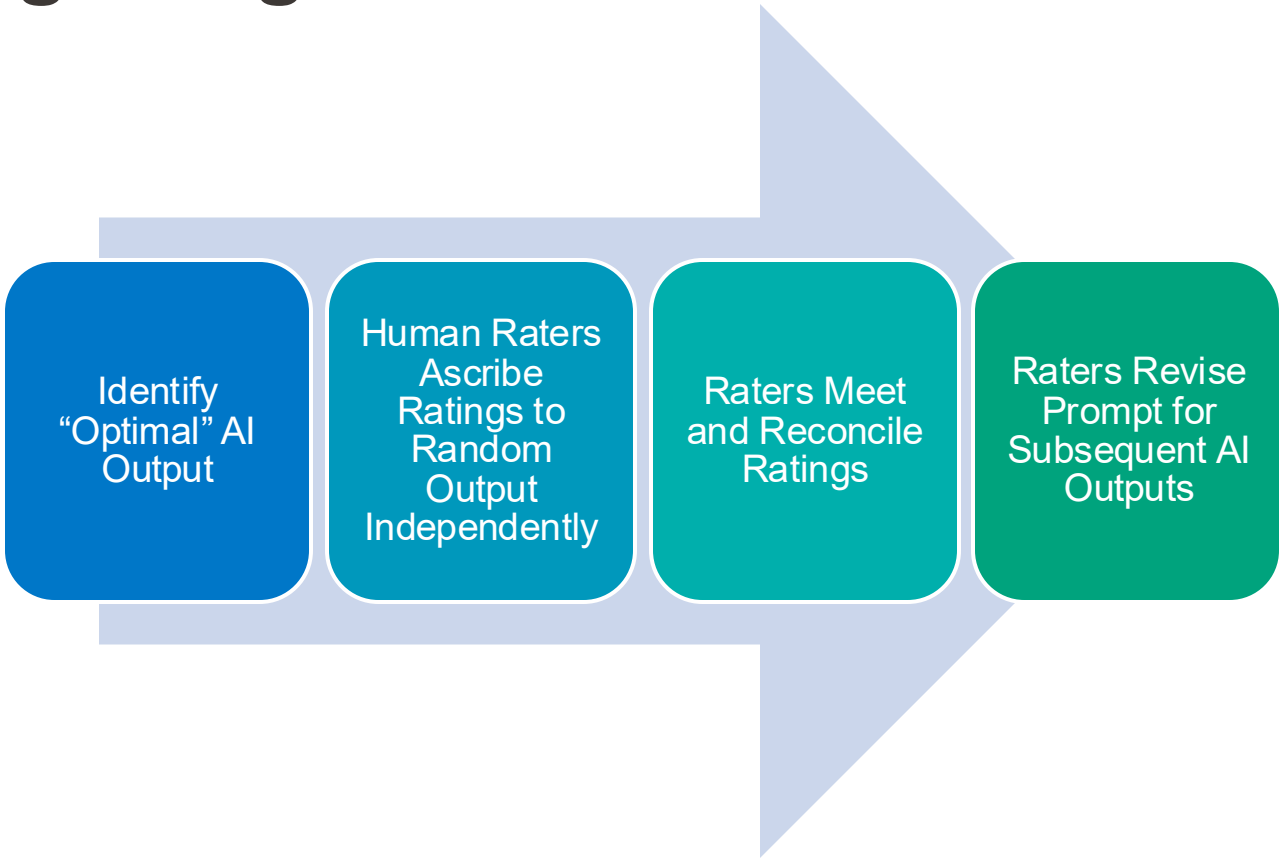
Create explicit code category saturation rule prohibiting more than 10% of codes from comprising any one category

Example:

Guidelines:
IMPORTANT:
Categories within the final “Assigned Category” list MAY NOT CONTAIN more than 10% more of terms from “Original Terms” than any other category in “Assigned Category”

> 90% of codes did not help us to discern meaning from code to code.

Decision #4: Augment Human-in-the-Loop by Integrating More Stakeholders



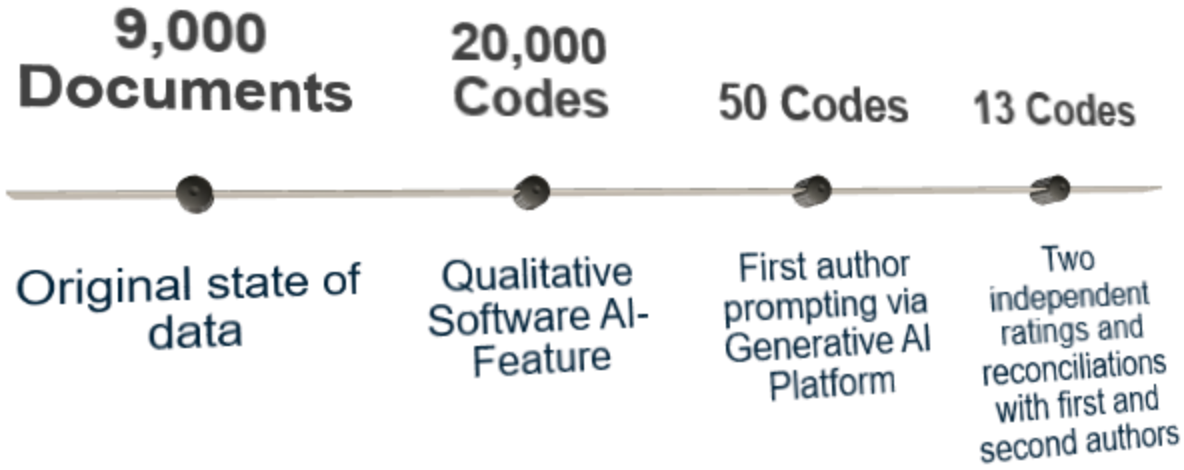
Decision #4: Augment Human-in-the-Loop by Integrating More Stakeholders

Original Code	Generated Code Category	SI Rating	MMC Rating	Final Category
Cognitive Switch	Other	?	Clinical Reasoning	Clinical Reasoning
Collaboration	Teamwork	Yes	Yes	Teamwork
Colleagues	challenges	Teamwork	Communication an/or Teamwork	Teamwork
Comfort	comfort	Quality of Care?	Growth?	Student Traits/Generic Skills
Comfort Zone	comfort	Feedback and Improvement?	Growth?	Feedback and Improvement
Commitment	encouragement	Motivation	Generic Skills	Student Traits/Generic Skills
Common Conditions	recognition	ddx	Medical Knowledge	Knowledge
Common Diagnoses	comfort	ddx	Medical Knowledge	Knowledge
Daily Reading	reading	Feedback and Improvement?	Study/Learning Practices	Study/Learning Practices

Decision #4: Augment Human-in-the-Loop by Integrating More Stakeholders

Original Code	Generated Category	Either Author Agree with AI?	SI Rating	MMC Rating	Authors Agree?	Final Category
Independence	Leadership	Disagree	Student Traits	Student Traits	Agree	Student Traits
Teamwork	Teamwork	Agree	Teamwork	Teamwork	Agree	Student Traits
Attentiveness	relationship building	Disagree	attention detail	genuine interest	Disagree	Attentiveness
Clinical Involvement	clinical reasoning	Disagree	patient care	Professionalism	Disagree	Engagement
Consistency	risk taking	Disagree	Student Traits	Professionalism	Disagree	Professionalism
Development	Feedback and Improvement	Agree	Feedback and Improvement	Study/Learning Practice	Disagree	Learning and Development
Examination Practice	Clinical Skills and Behaviors	Disagree	Clinical Skills and Behaviors	Study/Learning Practice	Disagree	Clinical Skills and Behaviors
General Medicine Standards	Clinical Skills and Behaviors	Disagree	Clinical Knowledge	Clinical Knowledge	Agree	Clinical Knowledge

Result


9,000 Documents
20,000 Codes
50 Codes
13 Codes
Original state of data
Qualitative Software AI-Feature
First author prompting via Generative AI Platform
Two independent ratings and reconciliations with first and second authors

Prospective Value of Process

Manual Coding (all files)

$$\frac{35\ files}{3\ days} = \frac{9000\ files}{\mathbf{772\ days\ (2.1\ years)}}$$

AI-Assisted Coding

$$\frac{35\ files}{3\ days} = \frac{9000\ files}{\mathbf{60\ days}}$$

Manual Coding (10% files)

$$\frac{35\ files}{3\ days} = \frac{1000\ files}{\mathbf{86\ days}}$$

Additional Time Savings for Future Work:

- Task Similarity
- Software Bandwidth
- Clear Instructions/Guidelines for Raters

What qualitative analysis challenge in your work feels most difficult to scale well, and how are you currently addressing it?

Study Findings

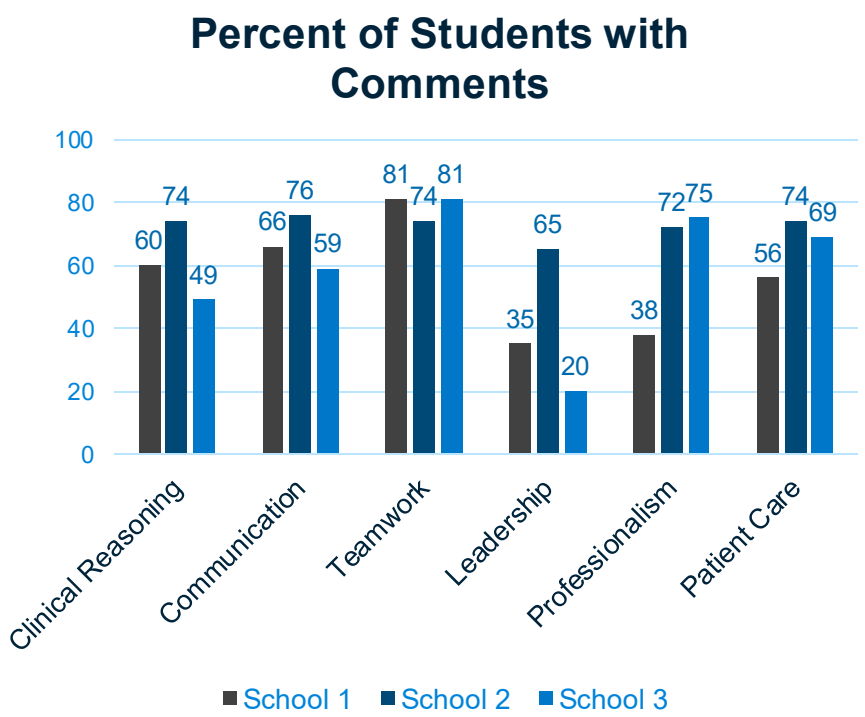
Themes and Valence

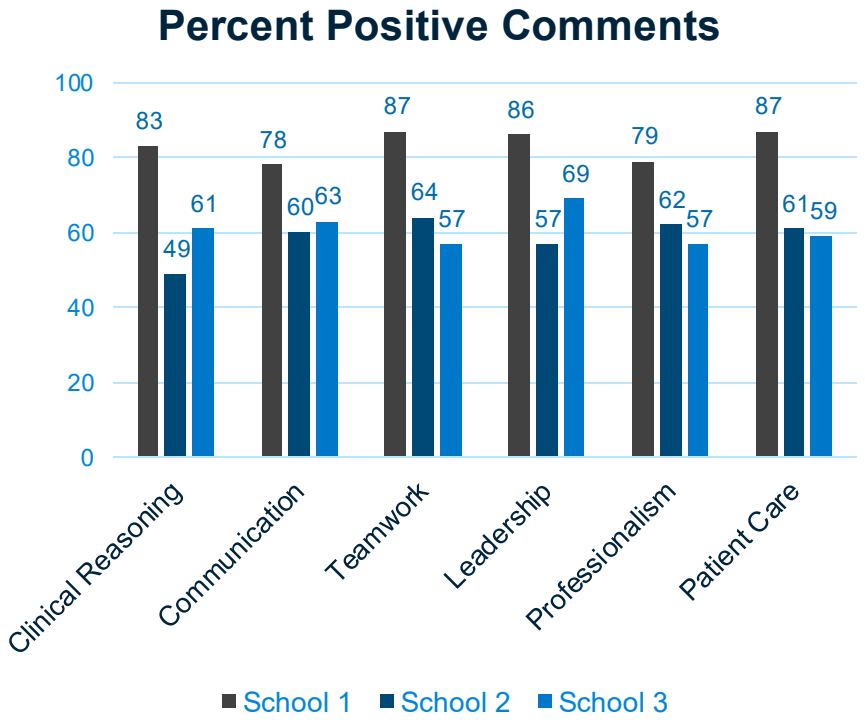
Competency Domains			
Communication	922 (66%)	Negative Neutral Positive	12 (3%) 63 (14%) 368 (83%)
Clinical Reasoning	839 (60%)	Negative Neutral Positive	9 (4%) 45 (18%) 192 (78%)
Teamwork	1,127 (81%)	Negative Neutral Positive	13 (2%) 89 (11%) 705 (87%)
Professionalism	538 (38%)	Negative Neutral Positive	1 (1%) 15 (16%) 79 (83%)
Leadership	491 (35%)	Negative Neutral Positive	0 (0%) 1 (13%) 7 (88%)
Patient Care	784 (56%)	Negative Neutral Positive	5 (2%) 23 (11%) 186 (87%)

Themes and Valence

Learning Approaches			
Motivation	1,259 (90%)	Negative Neutral Positive	32 (4%) 111 (12%) 756 (84%)
Growth Mindset	1,063 (76%)	Negative Neutral Positive	24 (4%) 114 (19%) 458 (77%)
Feedback	780 (56%)	Negative Neutral Positive	16 (6%) 33 (13%) 200 (80%)
Developmental Progress	1,007 (72%)	Negative Neutral Positive	40 (7%) 153 (26%) 406 (68%)
Self-Directed Learning	869 (62%)	Negative Neutral Positive	7 (3%) 31 (12%) 225 (86%)

Qualitative Data Comparisons

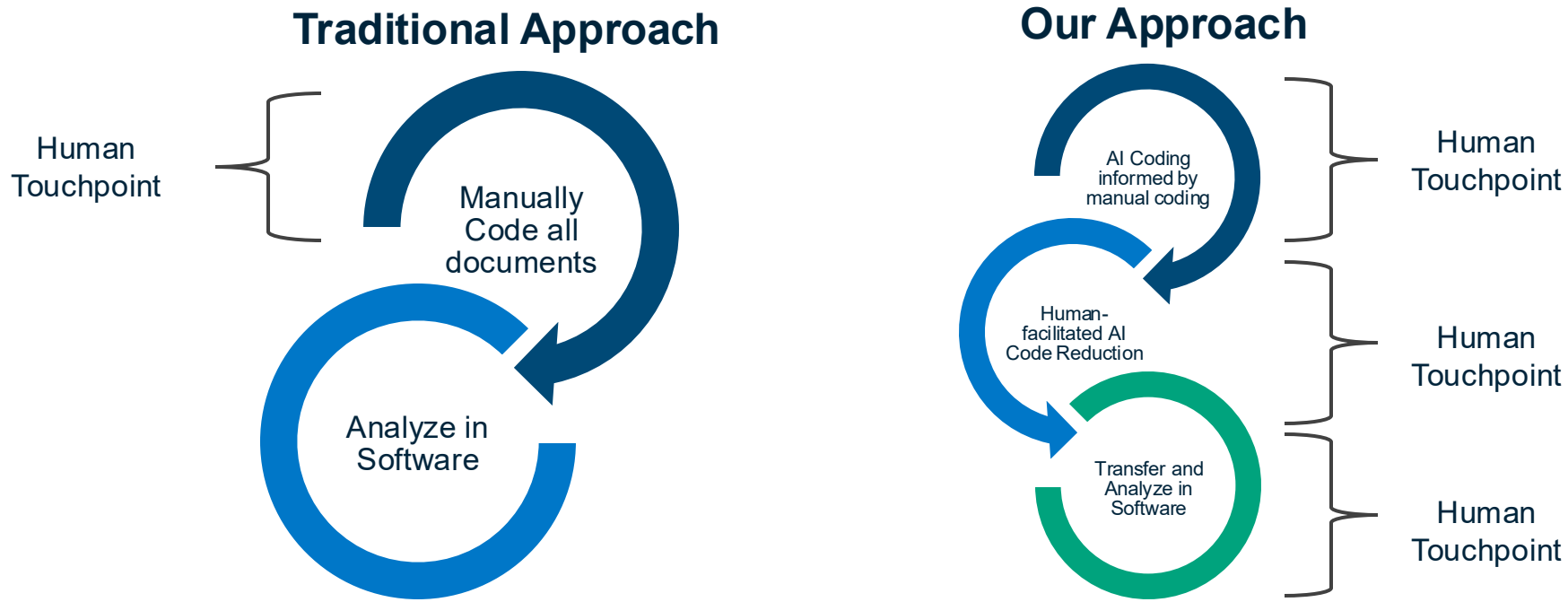




Negative Comments were between 2 and 5%

Lessons Learned and Next Steps

Lesson #1: AI did not eliminate completely work, but redistributed it



Next Step: *Explore Qualitatively Coding Employed Exclusively via Generative AI Platform*

Lesson #2: Human-in-the-Loop is a Component of Assessment Methodology, not a final check



Verifying accuracy

AND



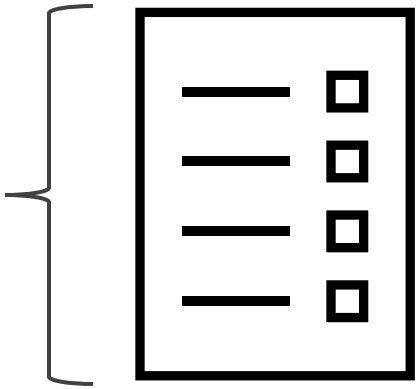
Preserving Validity and Ethics

Next Step: *Explore Best Practices for Replicability and Stronger Interrater and AI Agreement*

Lesson #3: Prompt Quality Matters (and so might other things)

Most effective prompts included:

- Multiple specific examples
- Explicit instructions
- Formatting Guidelines



Different platforms (e.g., OpenAI, Anthropic)
Different models (e.g., GPT 4.o, Sonnet, "reasoning" models)
Variation in Model Parameters (e.g., temperature)

***Next steps** might include exploring…*

When one AI approach falls short, another may make the task possible.

Works Referenced
Resources
Slides

We'd love to hear your thoughts/questions!

